

CHAPTER 4

4-1. A pole at s transforms into a pole at $z = e^{sT}$.

(a) $s = -a + jb$, $a > 0$

$$|z| = |e^{(-a+jb)T}| = |e^{-aT} e^{jbT}| = \underline{|e^{-aT}| < 1}$$

(b) $s = j\omega \Rightarrow z = e^{j\omega T} = \underline{1/\omega T}$

(c) $s = a + jb$, $a > 0$

$$|z| = |e^{(a+jb)T}| = |e^{aT} e^{jbT}| = \underline{|e^{aT}| > 1}$$

4-2. (a) $s = 1$, $\therefore z = e^T = e^{0.05} = \underline{1.051}$

$s = -1$, $\therefore z = e^{-T} = e^{-0.05} = \underline{0.9512}$

(b) A pole at s transforms into a pole at $z = e^{sT} = \underline{e^{0.05s}}$

(c) $E(s) = \frac{s+2}{(s-1)(s+1)} = \frac{3/2}{s-1} + \frac{-1/2}{s+1}$

$$\therefore E(z) = \frac{1.5z}{z-1.051} + \frac{-0.5z}{z-0.9512} = \frac{z^2 - 0.9013z}{(z-1.051)(z-0.9512)}$$

(d) $s = -2$; $z = \underline{0.9013}$

(e) No - both poles and zeros of $E(s)$ determine the zeros of $E(z)$.

4-3. (a) $E(z) = \frac{20(e^{-0.2} - e^{-0.5})z}{3(z - e^{-0.2})(z - e^{-0.5})} = \frac{1.415z}{(z - 0.8187)(z - 0.6065)}$

$$\omega_s = 2\pi/T = \underline{20\pi}$$

$E(s)$: $s = -2, -5$

$E^*(s)$: $s = -2 \pm jk20\pi$, $-5 \pm jk20\pi$

$E(z)$: $z = 0.8187, 0.6065$

(b) $E(z) = \frac{5z(1 - e^{-0.1})}{(z-1)(z - e^{-0.1})} = \frac{0.4758z}{(z-1)(z - 0.9048)}$

4-3.(b) poles: $E(s): s = 0, -1$
 $E^*(s): s = \pm jk20\pi, -1 \pm jk20\pi$
 $E(z): z = 1, 0.9048$

(c) $E(s) = \frac{s+2}{s(s+1)} = \frac{2}{s} + \frac{-1}{s+1}$
 $\therefore E(z) = \frac{2z}{z-1} + \frac{-z}{z-0.9048} = \frac{z(z-0.8096)}{(z-1)(z-0.9048)}$
poles: Same as (b),

(d) $E(s) = \frac{s+2}{s^2(s+1)} = \frac{2}{s^2} + \frac{-1}{s} + \frac{1}{s+1}$
 $\therefore E(z) = \frac{0.2z}{(z-1)^2} + \frac{-z}{z-1} + \frac{z}{z-0.9048} = \frac{0.1048z^2 - 0.0858z}{(z-1)^2(z-0.9048)}$
poles: Same as (b)

(e) $E(s) = \frac{s^2+5s+6}{s(s+4)(s+5)} = \frac{0.3}{s} + \frac{-0.5}{s+4} + \frac{1.2}{s+5}$
 $\therefore E(z) = \frac{0.3z}{z-1} - \frac{0.5z}{z-0.6703} + \frac{1.2z}{z-0.6065}$
 $= \frac{z^3 - 1.5841z^2 + 0.6231z}{(z-1)(z-0.6703)(z-0.6065)}$
poles: $E(s): s = 0, -4, -5$

$E^*(s): s = \pm jk20\pi, -4 \pm jk20\pi, -5 \pm jk20\pi$
 $E(z): z = 1, 0.6703, 0.6065$

(f) $E(s) = \frac{2}{(s+1)^2 + 2^2} \therefore$
 $\therefore E(z) = \frac{2}{z} \left[\frac{ze^{-0.1} \sin 0.2}{z^2 - 2ze^{-0.1} \cos 0.2 + e^{-0.2}} \right] = \frac{0.1798z}{z^2 - 1.7736z + 0.8187}$

poles: $E(s): s = -1 \pm j2 = 2.236 \angle \pm 116.6^\circ$
 $E^*(s): s = -1 \pm j2 \pm jk20\pi = -1 \pm j(2 + k20\pi)$
 $E(z): z = 0.8868 \pm j0.1798 = 0.9048 \angle \pm 11.46^\circ$

(g) `num=[0 1 5 6];`
`den=[1 9 20 0];`
`[r,p,k]=residue(num,den)`

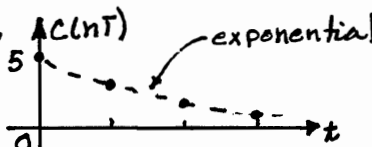
4-4. (a) $E(z) = \frac{(z^2-1)}{z^4} \cdot \left[\frac{1}{s(s+1)} \right] = \frac{z^2}{z^4} \cdot \frac{z(1-e^{-0.5})}{(z-1)(z-e^{-0.5})}$
 $= \frac{0.3935(z+1)}{z^3(z-0.6065)}$

$$4-4. (b) E(z) = (1-z^{-1}) \mathcal{Z} \left[\frac{0.5s+1}{0.5s(s+0.25)} \right] = \frac{z-1}{z} \mathcal{Z} \left[\frac{8}{s} + \frac{-7}{s+0.25} \right]$$

$$= \frac{z-1}{z} \left[\frac{8z}{z-1} - \frac{7z}{z-0.9344} \right] = \frac{z-0.5152}{z-0.9344}$$

$$4-5. (a) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{5s}{s(s+0.1)} \right] = \frac{z-1}{z} \times \frac{5z}{z-e^{-0.2}} = \frac{5(z-1)}{z-0.8187}$$

$$Y(z) = \left(\frac{z}{z-1} \right) \frac{5(z-1)}{z-0.8187} = \frac{5z}{z-0.8187}$$

$$\therefore c(nT) = \underline{5(0.8187)^n}$$


$$(b) M(s) = \frac{1}{s}$$

$$\therefore Y(s) = \frac{5s}{s(s+0.1)} = \frac{5}{s+0.1}, \therefore y(t) = \underline{5e^{-0.1t}}$$

$$\therefore y(nT) = y(2n) = 5e^{-0.1(2n)} = 5(e^{-0.2})^n = \underline{5(0.8187)^n}$$

$$(c) \text{dc gain} = G_p(s) \Big|_{s=0} = \frac{5(0)}{0+0.1} = \underline{0}$$

$$\text{dc gain} = G(z) \Big|_{z=1} = \frac{5(1-1)}{1-0.8187} = \underline{0}$$

$$(d) \text{Yes} - c_{ss}(kT) = (1)(\text{dc gain}) = 0 \text{ from (a)}, \therefore \text{dc gain} = 0$$

$$c_{ss}(t) = (1)(\text{dc gain}) = 0 \text{ from (b)}, \therefore \text{dc gain} = 0$$

$$4-6. (a) (i) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{5}{s(s+1)(s+2)} \right] = \frac{z-1}{z} \mathcal{Z} \left[\frac{2.5}{s} + \frac{-5}{s+1} + \frac{2.5}{s+2} \right]$$

$$= \frac{z-1}{z} \left[\frac{2.5z}{z-1} - \frac{5z}{z-0.9048} + \frac{2.5z}{z-0.8187} \right]$$

$$= \frac{z-1}{z} \left[\frac{0.0227z^2 + 0.0205z}{(z-1)(z-0.9048)(z-0.8187)} \right] = \frac{0.0227z + 0.0205}{(z-0.9048)(z-0.8187)}$$

$$\therefore Y(z) = \frac{0.0227z + 0.0205}{(z-1)(z-0.9048)(z-0.8187)} = \frac{2.5}{z-1} + \frac{-5}{z-0.9048} + \frac{2.5}{z-0.8187}$$

$$\therefore c(kT) = \underline{2.5 - 5(0.9048)^n + 2.5(0.8187)^n}$$

$$(b) M(s) = \frac{1}{s}, \therefore Y(s) = \frac{5}{s(s+1)(s+2)} = \frac{2.5}{s} - \frac{5}{s+1} + \frac{2.5}{s+2}$$

$$\therefore y(t) = 2.5 - 5e^{-t} + 2.5e^{-2t}$$

$$\therefore y(nT) = 2.5 - 5(0.9048)^n + 2.5(0.8187)^n$$

$$4-6. (c) \text{ dc gain} = G_p(s) \big|_{s=0} = \underline{2.5}$$

$$\text{dc gain} = G(z) \big|_{z=1} = \frac{0.0227 + 0.0225}{(1-0.9048)(1-0.8187)} = \underline{2.5}$$

$$(d) \text{ Yes} - C_{ss}(kT) = (1)(\text{dc gain}) = \underline{2.5}, \text{ from (a)}; \therefore \text{dc gain} = \underline{2.5}$$

$$C_{ss}(t) = (1)(\text{dc gain}) = \underline{2.5}, \text{ from (b)}; \therefore \text{dc gain} = \underline{2.5}$$

$$\begin{aligned} (b) (a) G(z) &= \frac{z-1}{z} \mathcal{Z} \left[\frac{5}{s[(s+1)^2+1]} \right] = \frac{z-1}{z} \mathcal{Z} \left[\frac{2.5}{s} + \frac{-2.5(s+1)}{(s+1)^2+1} + \frac{-2.5}{(s+1)^2+1} \right] \\ &= \frac{z-1}{z} \left[\frac{2.5z}{z-1} + \frac{-2.5z^2+2.251z}{z^2-1.801z+0.8187} + \frac{-0.2258}{z^2-1.801z+0.8187} \right] \\ &= \frac{0.02337z + 0.02187}{z^2 - 1.801z + 0.8187} \Rightarrow \text{poles: } z = 0.9048 \angle \pm 5.73^\circ \\ &= 0.900 \pm j0.0903 \end{aligned}$$

$$\begin{aligned} \therefore Y(z) &= \frac{0.02337z + 0.02187}{z(z-1)(z-0.900-j0.0903)(z-0.900+j0.0903)} \\ &= \frac{2.50}{z-1} + \frac{k_1}{z-0.900-j0.0903} + \frac{k_1^*}{z-0.900+j0.0903} \end{aligned}$$

$$\begin{aligned} k_1 &= \frac{0.02337(0.900+j0.0903)+0.02187}{(0.900+j0.0903-1)(2 \times 0.0903)} \\ &= \frac{0.04245 \angle 2.82^\circ}{(0.1347 \angle 132.9^\circ)(0.1806 \angle 90^\circ)} = 1.766 \angle -225.9^\circ \end{aligned}$$

$$\begin{aligned} \text{From (2-29), (2-30): } aT &= \ln(0.9048) = -1 \Rightarrow \underline{a = -1} \\ bT &= 5.73^\circ = 0.1 \text{ rad} \Rightarrow \underline{b = 1} \end{aligned}$$

$$A = 2|k_1| = \underline{3.531}, \quad \Theta = \underline{134.1^\circ}$$

$$\therefore y(nT) = \underline{2.5 + 3.531(0.9048)^n \cos(0.1n + 134.1^\circ)}$$

$$(b) M(s) = \frac{1}{s}$$

$$\therefore Y(s) = \frac{5}{s(s^2+2s+2)} = \frac{2.5}{s} + \frac{-2.5(s+1)}{(s+1)^2+1} + \frac{-2.5}{(s+1)^2+1}$$

$$\begin{aligned} \therefore y(t) &= 2.5 - 2.5(e^{-t} \cos t + e^{-t} \sin t) \\ &= 2.5 + 3.535 e^{-t} \cos(t + 135^\circ) \end{aligned}$$

$$\begin{aligned} \therefore y(nT) &= 2.5 + 3.535 e^{-0.1n} \cos(0.1n + 135^\circ) \\ &= \underline{2.5 + 3.535(0.9048)^n \cos(0.1n + 135^\circ)} \end{aligned}$$

$$4-6.(b) \quad (c) \quad G_p(s)|_{s=0} = \frac{5}{2} = \underline{2.5}$$

$$G(z)|_{z=1} = \frac{0.02337 + 0.02187}{1 - 1.800 + 0.8187} \approx \underline{2.5}$$

(d) Same as (d) of part (a),

$$4-7.(a) \quad \text{dc gain} = G(z)|_{z=1} = 0, \therefore G(z) \text{ has a zero at } z=1.$$

$$(b) \quad \text{dc gain} = G(z)|_{z=1} = G_p(s)|_{s=0}, \therefore G_p(s) \text{ has a zero at } s=0.$$

$$\therefore \text{let } G_p(s) = s G_{p1}(s)$$

$$G(z) = \mathcal{Z} \left\{ \left(\frac{1-e^{-Ts}}{s} \right) (s G_{p1}(s)) \right\} = \frac{z-1}{z} \mathcal{Z} [G_{p1}(s)]$$

$\therefore G(z)$ has a zero at $z=1$.

$$(c) \quad \text{Note: } \mathcal{Z}[1-e^{-Ts}] = \frac{z-1}{z}. \therefore \text{this term cancels the pole at } z=1.$$

$$(d) \quad \text{dc gain} = \lim_{z \rightarrow 1} G(z) \therefore G(z) \text{ has a pole at } z=1.$$

$$(e) \quad G_p(s) \text{ has a pole at } s=0. \therefore \text{let } G_p(s) = \frac{1}{s} G_{p1}(s)$$

$$\therefore G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{1}{s^2} G_{p1}(s) \right] = \frac{z-1}{z} \frac{N(z)}{(z-1)^2 D_1(z)} = \frac{N(z)}{z(z-1)D_1(z)}$$

$$4-8. \quad C(z) = \mathcal{Z} \left[\frac{1}{s(s+0.5)} \right] \frac{z-1}{z} \mathcal{Z} \left[\frac{3}{(s+1)(s+2)} \right]$$

$$\mathcal{Z} \left[\frac{1}{s(s+0.5)} \right] = \frac{1}{0.5} \frac{z(1-e^{-0.5})}{(z-1)(z-e^{-0.5})} = \frac{0.7869z}{(z-1)(z-0.6065)}$$

$$\frac{z-1}{z} \mathcal{Z} \left[\frac{3}{s(s+1)(s+2)} \right] = \frac{z-1}{z} \left[\frac{1.5}{s} + \frac{-3}{s+1} + \frac{1.5}{s+2} \right]$$

$$= \frac{z-1}{z} \left[\frac{1.5z}{z-1} + \frac{-3z}{z-0.3679} + \frac{1.5z}{z-0.1353} \right] = \frac{0.5990z + 0.2184}{(z-0.3679)(z-0.1353)}$$

$$\therefore \frac{C(z)}{z} = \frac{0.4714z + 0.1719}{(z-0.6065)(z-0.3679)(z-0.1353)(z-1)}$$

$$= \frac{-10.35}{z-0.6065} + \frac{9.845}{z-0.3679} + \frac{-2.487}{z-0.1353} + \frac{2.991}{z-1}$$

$$\therefore c(nT) = \underline{2.991 - 10.35(0.6065)^n + 9.845(0.3679)^n - 2.487(0.1353)^n}$$

$$4-9(a) \quad C_u(z) = \mathcal{Z}\left[\frac{1-e^{-Ts}}{s}\right] \mathcal{Z}\left[\frac{z}{s^2}\right] = \mathcal{Z}\left[\frac{1}{s^2}\right] = \frac{zTz}{(z-1)^2}$$

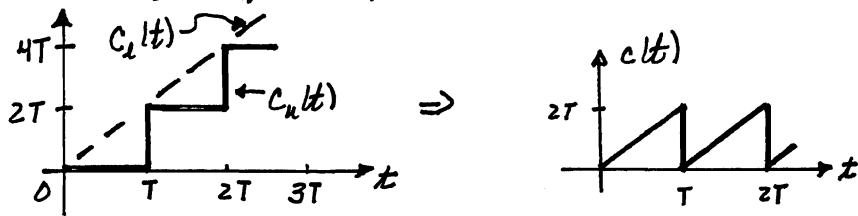
$$C_e(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{z}{s^2}\right] \mathcal{Z}\left[\frac{z}{z-1}\right] = \mathcal{Z}\left[\frac{1}{s^2}\right] = \frac{zTz}{(z-1)^2}$$

$$\therefore C(z) = C_e(z) - C_u(z) = 0$$

(b) No effect. The output of the sampler and data hold at the sampling instants is equal to the input.

(c) $C_u(t) = \text{output of data hold.}$

$C_e(t) = \text{integral of step function} = zt$



$$(d) \quad C_u(t) = C_e(t), \quad \therefore C(t) = 0$$

$$4-10.(a) \quad (i) \quad C(z) = G_3(z) D(z) E(z) + \overline{G_1 G_2} E(z)$$

$$(ii) \quad C(z) = \overline{G_1 G_3 G_4} E(z) + \overline{G_2 G_4} (z) \overline{G_1} E(z)$$

$$(iii) \quad C(z) = \overline{G_1 G_2 G_3} E(z) + G_3'(z) D(z) \overline{G_1} E(z); \quad G_3'(z) = \left(\frac{1-e^{-Ts}}{s}\right) G_3(s)$$

$$(b) \quad (i) \quad G_3(s)$$

$$(ii) \quad G_2(s)$$

$$(iii) \quad G_3'(s)$$

$$4-11.(a) \quad M(z) = (z - z^{-1}) E(z)$$

$$\therefore m(kT) = 2e(kT) - e[(k-1)T] = 2u(kT) - u[(k-1)T]$$

$$\therefore m(0) = 2, \quad m(k) = 1, \quad k \geq 1$$

$$(b) \quad c(kT) = 2c_1(kT) - c_1[(k-1)T]$$

$$(c) \quad c_1(kT) = (1 - e^{-kT}) u(kT)$$

$$c(kT) = 2(1 - e^{-kT}) u(kT) - u[(k-1)T] + e^{-(k-1)T} u[(k-1)T]$$

$$= \begin{cases} 0, & k=0 \\ 1 - (2 - e^{-T}) e^{-kT}, & k \geq 1 \end{cases}$$

4-11.(d) Example 4.3, $C_1(kT) = \frac{z(1-\epsilon^{-T})}{(z-1)(z-\epsilon^{-T})}$

$$\begin{aligned}\therefore C(z) &= z C_1(z) - z^{-1} C_1(z) = \frac{z^2(1-\epsilon^{-T})}{(z-1)(z-\epsilon^{-T})} - \frac{1-\epsilon^{-T}}{(z-1)(z-\epsilon^{-T})} \\ &= \frac{(z^2-1)(1-\epsilon^{-T})}{(z-1)(z-\epsilon^{-T})}\end{aligned}$$

4-12.(a) dc gain = $D(1) = \frac{4.5(1-0.9)}{1-0.85} = \underline{3}$

- (b) 1. Use hardware configuration of Figure P4-12.
2. Apply a unit step input voltage and let the output settle to a constant value.
3. Use an oscilloscope to measure the output voltage, which should be equal to 3V.

4-13.(a) $M(z) = -0.9z^{-1}M(z) + 0.2E(z)$

$$\therefore D(z) = \frac{M(z)}{E(z)} = \frac{0.2z}{z-0.9}$$

$$\begin{aligned}G(z) &= \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s(s+0.2)}\right] = \frac{z-1}{z} \left[\frac{1}{0.2} \frac{z(1-\epsilon^{-0.2})}{(z-1)(z-\epsilon^{-0.2})}\right] \\ &= \frac{0.9063}{z-0.8187}\end{aligned}$$

$$\therefore \frac{C(z)}{E(z)} = D(z) G(z) = \frac{0.1813z}{(z-0.9)(z-0.8187)}$$

(b) $D(1) G(1) = \frac{0.1813}{(1-0.9)(1-0.8187)} = \underline{10}$

(c) $D(1) = \frac{0.2}{1-0.9} = \underline{2}$; $G_p(0) = \frac{1}{0.2} = \underline{5}$

$$D(1) G_p(0) = (2)(5) = \underline{10}$$

(d) $C_{ss}(kT) = D(1) G(1) \times (1) = \underline{10}$

$$(e) C(z) = z \left[\frac{0.1813z}{(z-1)(z-0.9)(z-0.8187)} \right] = \frac{10}{z-1} + \frac{-20.07}{z-0.9} + \frac{10.07}{z-0.8187}$$

$$\therefore C(kT) = 10 - 20.07(0.9)^k + 10.07(0.8187)^k, \therefore C_{ss}(kT) = \underline{10}$$

4-13. (f) If the sum of the partial-fraction coefficients is not zero, then $C(z) = z C_1(z) = \frac{az^3 + \dots}{z^3 + \dots} = a + ()z^{-1} + ()z^{-2} \dots$

Hence $c(0)$ is not zero, and the plant responds instantly to a step input. This does not occur.

4-14. (a) $zM(z) - 0.995M(z) = 0.5(z - 0.98)E(z)$

$$\therefore D(z) = \frac{M(z)}{E(z)} = \frac{0.5(z - 0.98)}{z - 0.995}$$

$$G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{5}{s(s+1)(s+2)} \right] = \frac{0.0227z + 0.0205}{(z - 0.9048)(z - 0.8187)} \text{, from Prob 4-6}$$

$$\therefore \frac{C(z)}{E(z)} = D(z)G(z) = \frac{(z - 0.98)(0.01135z + 0.01025)}{(z - 0.995)(z - 0.9048)(z - 0.8187)}$$

(b) $D(1)G(1) = \frac{(1 - 0.98)(0.01135 + 0.01025)}{(1 - 0.995)(1 - 0.9048)(1 - 0.8187)} = \underline{5}$

(c) $D(1) = \frac{0.5(1 - 0.98)}{1 - 0.995} = \underline{2}$; $G_p(0) = \frac{5}{2} = \underline{2.5}$

$$\therefore D(1)G_p(0) = (2)(2.5) = \underline{5}$$

(d) $c_{ss}(kT) = D(1)G(1) \times (1) = \underline{5}$

(e) $\frac{C(z)}{z} = \frac{(z - 0.98)(0.01135z + 0.01025)}{(z - 1)(z - 0.995)(z - 0.9048)(z - 0.8187)} = C_1(z)$

$$= \frac{5}{z-1} + \frac{-4.064}{z-0.995} + \frac{-2.087}{z-0.9048} + \frac{1.145}{z-0.8187}$$

$$\therefore C(kT) = 5 - 4.064(0.995)^k - 2.087(0.9048)^k + 1.145(0.8187)^k$$

(f) See Problem 4-13(f)



(b) $\pm 10V$. Then the input motor voltage is in the range of $\pm 24V$, with $24V$ the rated motor voltage.

$$4-15.(c) \frac{\Theta_a(s)}{E_a(s)} = \frac{2}{s(0.5s+1)} = \frac{4}{s(s+2)}$$

$$(d) \frac{\Theta_a(z)}{M(z)} = \frac{z-1}{z} \mathcal{Z} \left[\frac{9.6}{s^2(s+2)} \right] = 9.6 \left(\frac{z-1}{z} \right) \left(\frac{1}{4} \right) \frac{z[0.2-1+0.8187]z+(1-0.8187-1/637)]}{(z-1)^2(z-0.8187)}$$

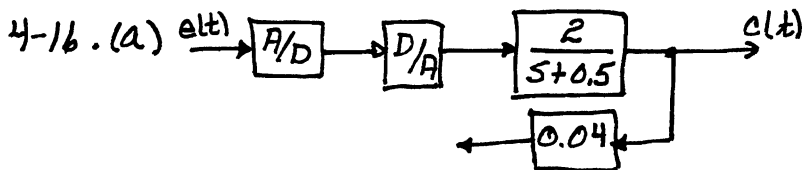
$$= \frac{0.04488z + 0.04206}{(z-1)(z-0.8187)}$$

$$(e) \Theta_a(s) = \frac{1}{s} \frac{9.6}{s(s+2)} = \frac{9.6}{s^2(s+2)} = \frac{4.8}{s^2} + \frac{-2.4}{s} + \frac{2.4}{s+2}$$

$$\therefore \Theta_a(t) = 4.8t - 2.4 + 2.4e^{-2t}$$

$$\therefore \Theta_{ass}(t) = 4.8t - 2.4, \quad \dot{\Theta}_{ass}(t) = 4.8 = \text{shaft velocity}$$

Constant voltage into a motor gives constant shaft velocity.



$$(b) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{2}{s(s+0.5)} \right] = \frac{z-1}{z} \frac{2}{0.5} \left[\frac{z(1-e^{-0.5T})}{(z-1)(z-e^{-0.5T})} \right]$$

$$= \frac{4(1-e^{-0.5T})}{z-e^{-0.5T}}$$

$$(c) G_p(0) = dc \text{ gain} = \frac{2}{0.5} = 4, \quad \therefore C_{ss}(kT) = 10(4) = 40^\circ C$$

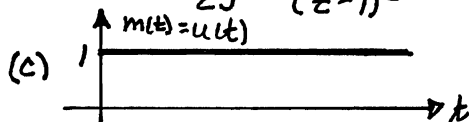
$$(d) G_d(0) = -\frac{2.5}{0.5} = -5, \quad \therefore \Delta C_{ss}(kT) = (-5)(1) = -5^\circ C$$

$$(e) C(s) = E^*(s) \left(\frac{1-e^{-Ts}}{s} \right) \left(\frac{2}{s+0.5} \right) - \frac{2.5}{s+0.5} D(s)$$

$$4-17. (a) G(z) = K \left(\frac{z-1}{z} \right) \mathcal{Z} \left[\frac{1}{Js^3} \right] = \frac{z-1}{z} \left(\frac{K}{J} \right) \frac{T^2 z(z+1)}{2(z-1)^3}$$

$$= \frac{KT^2(z+1)}{2J(z-1)^2}$$

$$(b) C(z) = \frac{KT^2}{2J} \frac{(z+1)z}{(z-1)^3}, \quad \therefore C(kT) = \frac{K}{J} \frac{(kT)^2}{2} = \frac{K}{2J} (kT)^2$$



$$4-17. (d) C(s) = \left(\frac{K}{s^2} \right) \frac{1}{s} = \frac{K}{s} \cdot \frac{1}{s^3}, \therefore c(t) = \underline{\frac{K}{J} \left(\frac{t^2}{2} \right)}$$

$$(e) \underline{c(kT) = \frac{K}{2J} (kT)^2}$$

$$\begin{aligned} 4-18. (a) K G(z) &= \frac{z-1}{z} K \mathcal{Z} \left[\frac{20}{s^2(s+6)} \right] \\ &= \left(\frac{z-1}{z} \right) \frac{20K}{36} \left[\frac{z(0.3-1+0.7408) + (1-0.7408-0.2222)}{(z-1)^2(z-0.7408)} \right] \\ &= K \left[\frac{0.02268z + 0.02052}{(z-1)(z-0.7408)} \right] \end{aligned}$$

$$(b) E(z) = 10$$

$$\therefore \frac{C(z)}{z} = \frac{10K(0.02268z + 0.02052)}{z(z-1)(z-0.7408)} = \frac{0.277K}{z} + \frac{1.67K}{z-1} + \frac{-1.94K}{z-0.7408}$$

$$c(kT) = 0.277K \delta(k) + 1.67K - 1.94K(0.7408)^k$$

$$\therefore \underline{C_{ss}(kT) = 1.67K}$$

$$(c) \text{ In (b), } C(z) = \frac{az + \dots}{z^2 + \dots} = az^{-1} + \dots = C(0) + C(1)z^{-1} + \dots$$

$$\therefore \underline{C(0) = 0}$$

$$\text{In } c(kT) \text{ in (b), } c(0) = K[0.277 + 1.67 - 1.94] \approx 0$$

(d) A constant voltage is applied to the motor. Thus the motor speed increases to a constant value, and the shaft angle $\theta(t)$ is a ramp voltage.

(e) From (d), the antenna rotates at a constant rpm.

4-19. From tables:

$$(a) E(s) = \frac{20}{(s+2)(s+5)} = \frac{20/3}{s+2} - \frac{20/3}{s+5}$$

$$\therefore E(z, m) = \frac{\frac{20}{3} e^{-2mT}}{z - e^{-2T}} - \frac{\frac{20}{3} e^{-5mT}}{z - e^{-5T}}$$

$$(b) E(s) = \frac{5}{s(s+1)} \Rightarrow E(z, m) = \frac{5}{z-1} - \frac{5e^{-mT}}{z - e^{-T}}$$

$$(c) E(s) = \frac{s+2}{s(s+1)} = \frac{2}{s} + \frac{-1}{s+1} \Rightarrow E(z, m) = \frac{2}{z-1} - \frac{e^{-mT}}{z - e^{-T}}$$

$$4-19. (d) E(s) = \frac{s+2}{s^2(s+1)} = \frac{2}{s^2} + \frac{-1}{s} + \frac{1}{s+1}, \text{ from Problem 4-3(d)}$$

$$E(z, m) = \mathcal{Z}\left[\frac{mT}{z-1} + \frac{T}{(z-1)^2}\right] = \frac{1}{z-1} + \frac{e^{-mT}}{z-e^{-T}}$$

$$(e) E(s) = \frac{s^2+5s+6}{s(s+4)(s+5)} = \frac{0.3}{s} + \frac{-0.5}{s+4} + \frac{1.2}{s+5}, \text{ from Problem 4-3(e)}$$

$$E(z, m) = \frac{0.3}{z-1} - \frac{0.5e^{-4mT}}{z-e^{-4T}} + \frac{1.2e^{-5mT}}{z-e^{-5T}}$$

$$(f) E(s) = \frac{2}{(s+1)^2+2^2}; E(z, m) = \frac{2}{z^2-2ze^{-T}\cos 2T + e^{-2T}} \left[\frac{e^{-mT}[ze^{j2mT}\sin(2mT) + e^{-T}\sin(2(1-m)T)]}{z^2-2ze^{-T}\cos 2T + e^{-2T}} \right]$$

4-20. Results of Problem 4-19 used.

$$(a) E(z, m)|_{m=0.7} = E(z, 0.7) = \frac{20}{3} \frac{e^{-1.4T}}{z-e^{-2T}} - \frac{20}{3} \frac{e^{-3.5T}}{z-e^{-5T}}$$

$$(b) E(z, 0.4) = \frac{5}{z-1} - \frac{5e^{-0.4T}}{z-e^{-T}}$$

$$(c) z^{-1}E(z, 0.9) = \frac{z}{z(z-1)} - \frac{e^{-0.9T}}{z(z-e^{-T})}$$

$$(d) E(z, 0.8) = 2 \left[\frac{0.8T}{z-1} + \frac{T}{(z-1)^2} \right] - \frac{1}{z-1} + \frac{e^{-0.8T}}{z-e^{-0.8T}}$$

$$(e) E(z, 0.7) = \frac{0.3}{z-1} - \frac{0.5e^{-2.8T}}{z-e^{-4T}} + \frac{1.2e^{-3.5T}}{z-e^{-5T}}$$

$$(f) E(z, 0.25) = \frac{1}{2} \left[\frac{e^{-0.25T}[ze^{j0.5T}\sin(0.5T) + e^{-T}\sin(1.5T)]}{z^2-2ze^{-T}\cos 2T + e^{-2T}} \right]$$

$$4-21. \frac{z-1}{z} \mathcal{Z}\left[\frac{G_p(s)}{s}\right] = \frac{2(z-1)}{z} \mathcal{Z}\left[\frac{e^{-3.33Ts}}{s(s+0.5)}\right] = \frac{2(z-1)}{z} z^{-3} \mathcal{Z}_m\left[\frac{1}{s(s+0.5)}\right]_{m=0.667}$$

$$= \frac{2(z-1)}{z^4(0.5)} \left[\frac{1}{z-1} - \frac{e^{-(0.5)(0.667)(0.6)}}{z-e^{-(0.5)(0.6)}} \right] = \frac{4(z-1)}{z^4} \left[\frac{0.1814z+0.0778}{(z-1)(z-0.7408)} \right]$$

$$= \frac{0.7256z+0.3112}{z^4(z-0.7408)}$$

$$(a) \text{ From Problem 4-16, } G(z) = \frac{4(1-e^{-0.3})}{z-e^{-0.3}} = \frac{1.037}{z-0.7408}$$

$$\therefore C(z) = z \left[\frac{1.037}{(z-1)(z-0.7408)} \right] = \frac{4}{z-1} - \frac{4}{z-0.7408}$$

$$\therefore c(kT) = 4 - 4(0.7408)^k$$

$$4-21.(b) \quad C(z) = \frac{(0.7256z - 0.3112)z}{z^4(z-1)(z-0.7408)} = z^{-4}C_1(z)$$

$$\therefore \frac{C_1(z)}{z} = \frac{0.7256z - 0.3112}{(z-1)(z-0.7408)} = \frac{4}{z-1} + \frac{-3.2744}{z-0.7408}$$

$$\therefore C_1(kT) = 4 - 3.2744(0.7408)^k$$

$$C(kT) = [4 - 3.2744(0.7408)^{(k-4)}]u(k-4)$$

$$(c) \quad C(s) = \frac{2}{s(s+0.5)} = \frac{4}{s} - \frac{4}{s+0.5} \Rightarrow C(t) = 4(1 - e^{-0.5t})$$

$$(d) \quad C(t-2) = 4[1 - e^{-0.5(t-2)}]u(t-2)$$

$$(e) \quad C(kT) = 4[1 - e^{-(0.5)(0.6)k}] = 4[1 - (0.7408)^k]$$

$$\begin{aligned} (f) \quad C(kT) &= 4[1 - e^{1-e^{-0.5(kT)}}]u(k-4) \\ &= [4 - 4e^{1-e^{-0.5(k-4)}}e^{-1/2}]u(k-4) \\ &= [4 - 3.2749(0.7408)^{(k-4)}]u(k-4) \end{aligned}$$

$$4-22.(a) \quad D(z) = 1.2 + \frac{0.1z}{z-1} = \frac{1.3z-1}{z-1}; \text{ From Prob. 4-21(a): } G(z) = \frac{1.037}{z-0.7408}$$

$$\therefore \frac{C(z)}{z} = \frac{1.037(1.3z-1)}{(z-1)^2(z-0.7408)} = \frac{1.20}{(z-1)^2} + \frac{0.5705}{z-1} + \frac{-0.5705}{z-0.7408}$$

$$\therefore C(z) = \frac{2(0.6z)}{(z-1)^2} + \frac{0.5705z}{z-1} + \frac{-0.5705z}{z-0.7408}$$

$$\therefore C(kT) = \underline{2kT + 0.5705[1 - 0.7408^k]}$$

(b) The temperature increases without limit - Not physically realizable.

$$(c) \quad \frac{M(z)}{z} = \frac{1.3z-1}{(z-1)^2} = \frac{1.3(z-1)}{(z-1)^2} + \frac{0.3}{(z-1)^2} = \frac{0.3}{(z-1)^2} + \frac{1.3}{z-1}$$

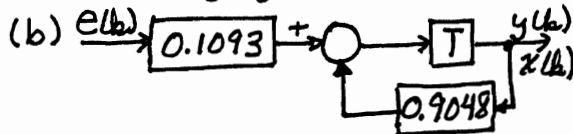
$$\therefore m(kT) = \underline{kT + 1.3}$$

(d) The value will reach full open. Then $m(kT)$ is effectively constant and the temperature will settle to a constant value.

$$4-23.(a) (s+0.05)Y(s) = 0.1M(s), \therefore G_p(s) = \frac{0.1}{s+0.05}$$

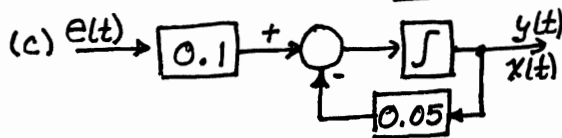
$$G(z) = \frac{Y(z)}{E(z)} = \frac{z-1}{z} \mathcal{Z} \left[\frac{0.1}{s(s+0.05)} \right]$$

$$= \frac{z-1}{z} \left(\frac{0.1}{0.05} \right) \frac{z(1-e^{-(0.05)(2)})}{(z-1)(z-0.9048)} = \frac{0.1903}{z-0.9048}$$



$$x(k+1) = 0.9048x(k) + 0.1093e(k)$$

$$y(k) = x(k)$$



$$\dot{x}(t) = -0.05x(t) + 0.1e(t)$$

$$y(t) = x(t)$$

$$(d) (sI-A)^{-1} = \frac{1}{s+0.05}, \therefore \Phi(t) = e^{-0.05t}$$

$$\therefore A = \Phi(T) = 0.9048$$

$$B = B_c \int_0^T \Phi(\tau) d\tau = B_c \int_0^2 e^{-0.05\tau} d\tau$$

$$= \frac{0.1}{-0.05} e^{-0.05\tau} \Big|_0^2 = 2[1-0.9048] = 0.1903$$

$$\therefore x(k+1) = 0.9048x(k) + 0.1093e(k)$$

$$y(k) = x(k)$$

(e) Same as (b),

$$(f) G(z) = \frac{0.1093z^{-1}}{1-0.9048z^{-1}} = \frac{0.1093}{z-0.9048}$$

(g)

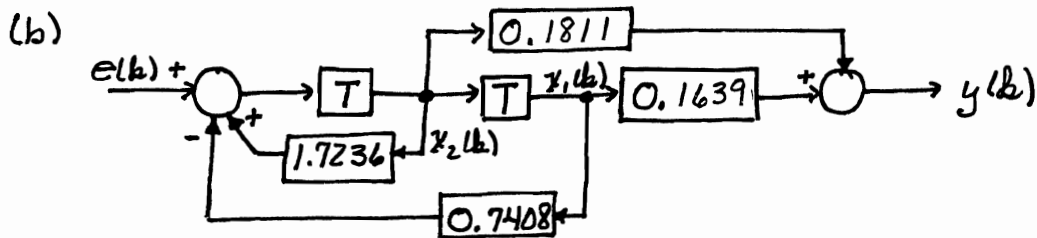
```
num=[0 0.1];
den=[1 0.05];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,2)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = -.05; Bc = 0.1;
[A,B] = c2d(Ac,Bc,2)
```

$$4-24.(a) [s^2+0.15s+0.005]Y(s) = 0.1M(s) \Rightarrow G_p(s) = \frac{0.1}{s^2+0.15s+0.005}$$

$$\therefore G(z) = \frac{Y(z)}{E(z)} = \frac{z-1}{z} \mathcal{Z} \left[\frac{0.1}{s(s+0.1)(s+0.05)} \right]$$

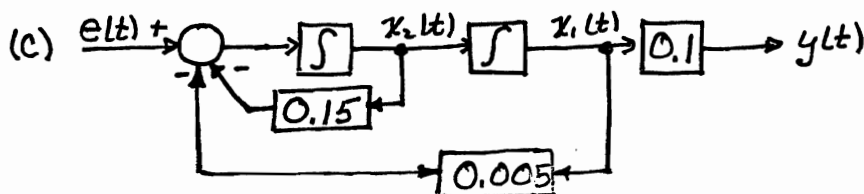
$$= \frac{z-1}{z} \mathcal{Z} \left[\frac{z}{s} + \frac{-2.5}{s+0.1} + \frac{0.5}{s+0.05} \right]$$

$$4-24.(a) \quad G(z) = \frac{z-1}{z} \left[\frac{2z}{z-1} + \frac{-2.5z}{z-0.8187} + \frac{0.5z}{z-0.9048} \right] = \frac{0.1811z + 0.1639}{z^2 - 1.7236z + 0.7408}$$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.7408 & 1.7236 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [0.1639 \quad 0.1811] \underline{x}(k)$$



$$\dot{\underline{x}}(t) = \begin{bmatrix} 0 & 1 \\ -0.005 & -0.15 \end{bmatrix} \underline{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(t)$$

$$y(t) = [0.1 \quad 0] \underline{x}(t)$$

(d) $sI - A = \begin{bmatrix} s & -1 \\ 0.005 & s + 0.15 \end{bmatrix}$, $\Delta = |sI - A| = s^2 + 0.15s + 0.005 = (s + 0.1)(s + 0.05)$

$$\therefore (sI - A)^{-1} = \begin{bmatrix} \frac{s + 0.15}{(s + 0.1)(s + 0.05)} & \frac{1}{(s + 0.1)(s + 0.05)} \\ \frac{-0.005}{(s + 0.1)(s + 0.05)} & \frac{s}{(s + 0.1)(s + 0.05)} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-1}{s + 0.1} + \frac{2}{s + 0.05} & \frac{-20}{s + 0.1} + \frac{20}{s + 0.05} \\ \frac{0.1}{s + 0.1} + \frac{-0.1}{s + 0.05} & \frac{2}{s + 0.1} + \frac{-1}{s + 0.05} \end{bmatrix} = \Phi(s)$$

$$\therefore \Phi(t) = \begin{bmatrix} -e^{-0.1t} + 2e^{-0.05t} & -20e^{-0.1t} + 20e^{-0.05t} \\ 0.1e^{-0.1t} - 0.1e^{-0.05t} & 2e^{-0.1t} - e^{-0.05t} \end{bmatrix}$$

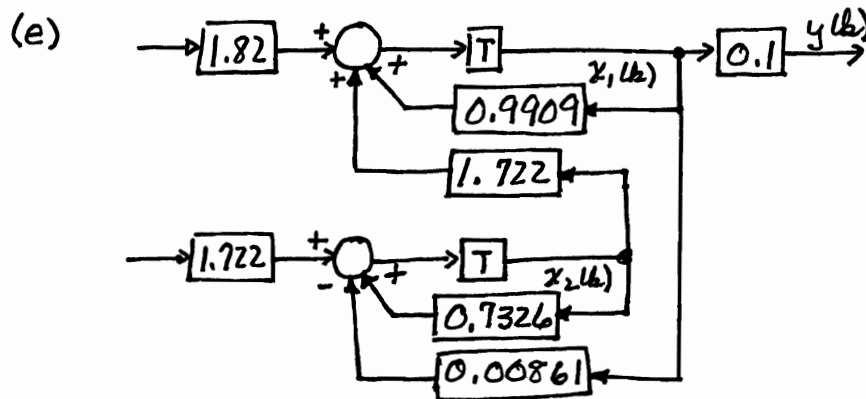
$$\therefore A = \Phi(2) = \begin{bmatrix} 0.9909 & 1.7220 \\ -0.00861 & 0.7326 \end{bmatrix}$$

$$\int_0^2 \Phi(t) dt = \begin{bmatrix} 10e^{-0.1t} - 40e^{-0.05t} & 200e^{-0.1t} - 400e^{-0.05t} \\ -e^{-0.1t} + 2e^{-0.05t} & -20e^{-0.1t} + 20e^{-0.05t} \end{bmatrix}_0^2$$

$$4-24(d) \therefore \int_0^2 \Phi(t) d\tau = \begin{bmatrix} 1 & 1.82 \\ 0 & 1.722 \end{bmatrix} \Rightarrow B = \left[\int_0^2 \Phi(t) dt \right] B_c = \begin{bmatrix} 1.82 \\ 1.722 \end{bmatrix}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0.9909 & 1.7220 \\ -0.00861 & 0.736 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1.82 \\ 1.722 \end{bmatrix} e(k)$$

$$y(k) = [0.1 \quad 0] \underline{x}(k)$$



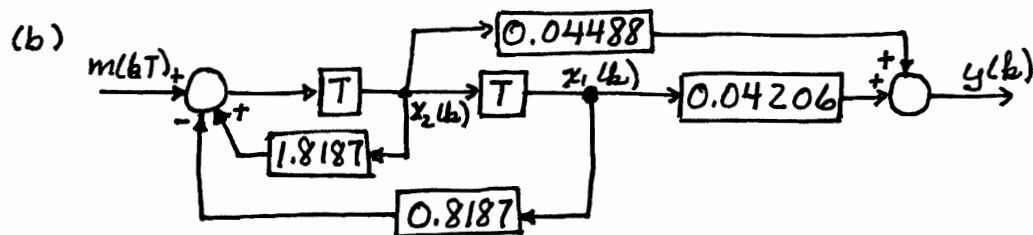
$$(f) Y(z) = \frac{0.182z^{-1}(1-0.7326z^{-1}) + 0.2965z^{-2}}{1-0.9909z^{-1}-0.7326z^{-1}+0.01483z^{-2}+0.7260z^{-2}} = \frac{0.182z + 0.1630}{z^2 - 1.7235z + 0.7408}$$

(g)

```

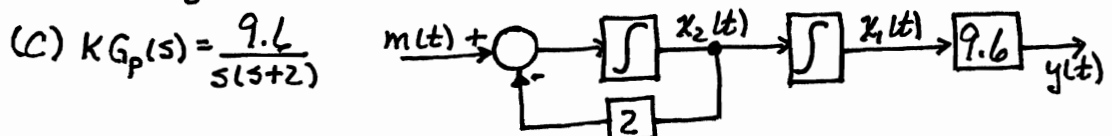
num=[0 0 0.1];
den=[1 0.15 0.005];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,2)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = [0 1; -.005 -.15]; Bc = [0;1];
[A,B] = c2d(Ac,Bc,2)
    
```

$$4-25, (a) \text{ From Problem P4-15: } G(z) = \frac{0.04488z + 0.04206}{z^2 - 1.8187z + 0.8187}$$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.8187 & 1.8187 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} m(k)$$

$$y(k) = [0.04206 \quad 0.04488] \underline{x}(k)$$



$$4-25.(c) \quad \dot{\underline{x}}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \underline{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} m(t)$$

$$y(t) = [9.6 \quad 0] \underline{x}(t)$$

$$(d) \quad sI - A_c = \begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}, \quad 1sI - A_c = \Delta = s^2 + 2s$$

$$(sI - A_c)^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix} \Rightarrow \Phi_c(t) = \begin{bmatrix} 1 & 0.5(1 - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix}$$

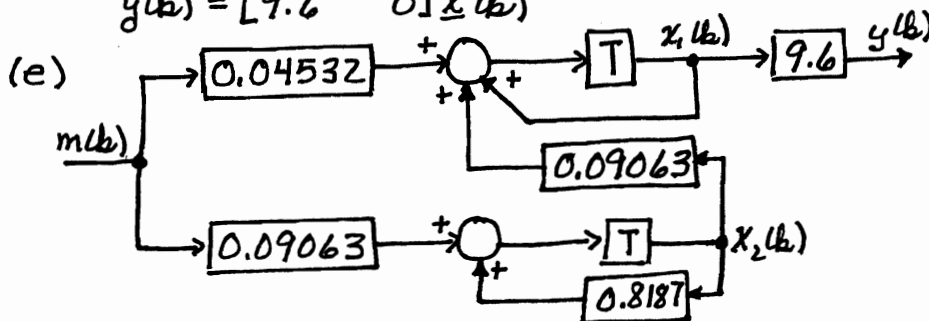
$$\therefore A = \Phi_c(T) = \begin{bmatrix} 1 & 0.09063 \\ 0 & 0.8187 \end{bmatrix}$$

$$\int_0^{0.1} \Phi_c(t) dt = \begin{bmatrix} t & 0.5t + 0.25e^{-2t} \\ 0 & -0.5e^{-2t} \end{bmatrix}_0^{0.1} = \begin{bmatrix} 0.1 & 0.04532 \\ 0 & 0.09063 \end{bmatrix}$$

$$B = \left[\int_0^{0.1} \Phi_c(t) dt \right] B_c = \begin{bmatrix} 0.04532 \\ 0.09063 \end{bmatrix}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 1 & 0.09063 \\ 0 & 0.8187 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.04532 \\ 0.09063 \end{bmatrix} m(k)$$

$$y(k) = [9.6 \quad 0] \underline{x}(k)$$



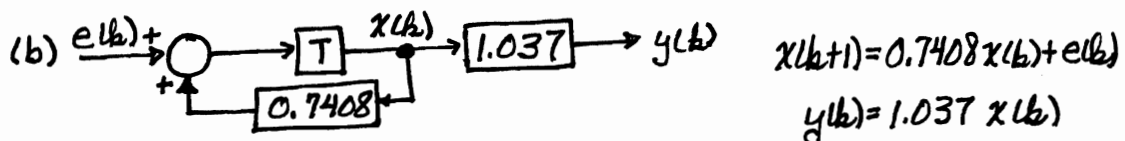
$$(f) \quad \frac{Y(z)}{M(z)} = \frac{0.04351z^{-1}(1 - 0.8187z^{-1}) + 0.07885z^{-2}}{1 - z^{-1} - 0.8187z^{-1} + 0.8187z^{-2}} = \frac{0.04351z + 0.04323}{z^2 - 1.8187z + 0.8187}$$

(g)

```

num=[0 0 9.61];
den=[1 2 0];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,0.1)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = [0 1; 0 -2]; Bc = [0;1];
[A,B] = c2d(Ac,Bc,0.1)
    
```

$$4-26.(a) \text{ From Problem 4-16 (a), } G(z) = \frac{1.037}{z - 0.7408} = \frac{C(z)}{E(z)}$$



4-26. (c) $G_p(s) = \frac{2}{s+0.5}$

$$\dot{x}(t) = -0.5x(t) + e(t)$$

$$y(t) = 2x(t)$$

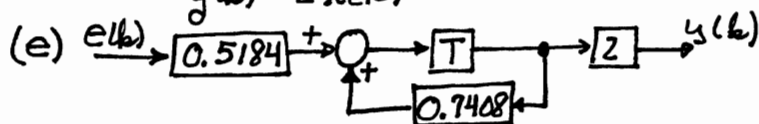
$$(d) \Phi_c(t) = \mathcal{S}^{-1}[(sI - A_c)^{-1}] = \mathcal{S}^{-1}\left[\frac{1}{s+0.5}\right] = e^{-0.5t}$$

$$A = \Phi_c(T) = e^{-0.5(0.6)} = 0.7408$$

$$B = B_c \int_0^T \Phi_c(t) dt = (1) \int_0^{0.3} e^{-0.5t} dt = -\frac{1}{0.5} e^{-0.5t} \Big|_0^{0.3} \\ = 2(1 - 0.7408) = 0.5184$$

$$\therefore x(k+1) = 0.7408x(k) + 0.5184e(k)$$

$$y(k) = 2x(k)$$



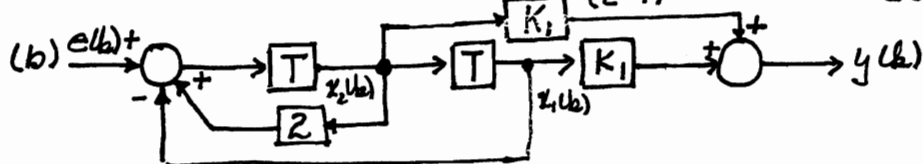
$$(f) \frac{Y(z)}{E(z)} = \frac{(0.5184)z^{-1}(z)}{1 - 0.7408z^{-1}} = \frac{1.037}{z - 0.7408}$$

(g)

```

num=[0 2];
den=[1 0.5];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,0.6)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = -.5; Bc = 1;
[A,B] = c2d(Ac,Bc,0.6)
    
```

4-27. (a) From Problem 4-17, $G(z) = K_1 \frac{z+1}{(z-1)^2}$ with $K_1 = \frac{KT^2}{2J} = \frac{K}{2J}$, $T=1$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [K_1 \quad K_1] \underline{x}(k)$$

(c) $G_p(s) = \frac{K}{Js^2}$

$$\dot{\underline{x}}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \underline{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(t)$$

$$y(t) = [K/J \quad 0] \underline{x}(t)$$

$$4-27.(d) \quad sI - A_c = \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}, \quad \Delta = |sI - A_c| = s^2, \quad \therefore (sI - A)^{-1} = \begin{bmatrix} 1/s & 1/s^2 \\ 0 & 1/s \end{bmatrix}$$

$$\therefore \Phi_c(t) = \mathcal{F}^{-1}[(sI - A_c)^{-1}] = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \Rightarrow A = \Phi_c(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\int_0^1 \Phi_c(t) dt = \begin{bmatrix} t & t^2/2 \\ 0 & t \end{bmatrix}_0^1 = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

$$\therefore B = \left[\int_0^1 \Phi_c(t) dt \right] B_c = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [K/J \quad 0] \underline{x}(k)$$



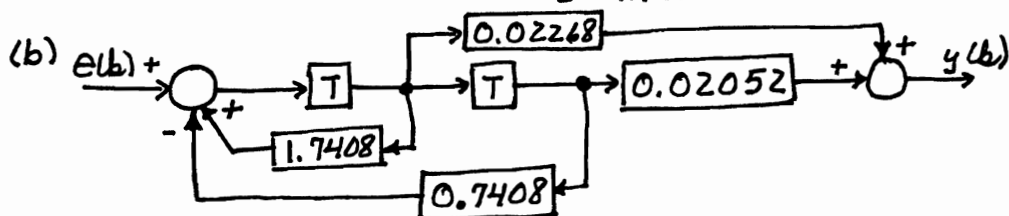
$$(f) \quad G(z) = \frac{0.5 \frac{K}{J} z^{-1} (1 - z^{-1}) + \frac{K}{J} z^{-2}}{1 - z^{-1} - z^{-1} + z^{-2}} = \frac{\frac{K}{J} (z+1)}{(z-1)^2}$$

(g)

```

num=[0 0 1];
den=[1 0 0];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,1)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = [0 1; 0 0]; Bc = [0; 1];
[A,B] = c2d(Ac,Bc,1)
    
```

$$4-28.(a) \text{ From Problem 4-18, } G(z) = \frac{0.02268z + 0.02052}{z^2 - 1.7408z + 0.7408}$$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.7408 & 1.7408 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [0.02052 \quad 0.02268] \underline{x}(k)$$

$$(c) \quad G_p(s) = \frac{20}{s(s+6)}$$

$$4-28.(c) \quad \dot{\underline{x}}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -6 \end{bmatrix} \underline{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} e(t)$$

$$y(t) = [20 \quad 0] \underline{x}(t)$$

$$(d) \quad sI - A_c = \begin{bmatrix} s & -1 \\ 0 & s+6 \end{bmatrix}, \quad \Delta = |sI - A_c| = s^2 + 6s$$

$$\therefore (sI - A_c)^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+6)} \\ 0 & \frac{1}{s+6} \end{bmatrix}, \quad \therefore \Phi_c(t) = \begin{bmatrix} 1 & \frac{1}{6}(1 - e^{-6t}) \\ 0 & e^{-6t} \end{bmatrix}$$

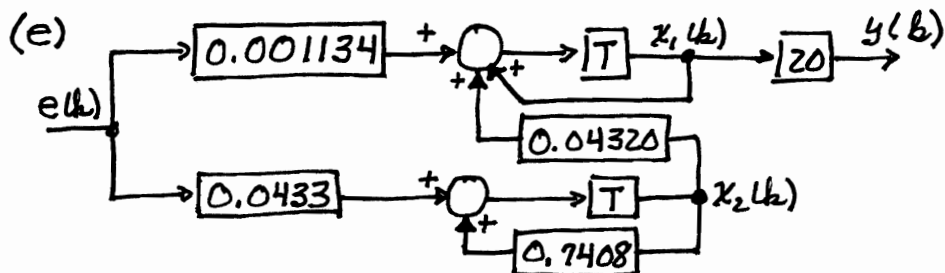
$$\therefore A = \Phi_c(T) = \begin{bmatrix} 1 & 0.04320 \\ 0 & 0.7408 \end{bmatrix}$$

$$\int_0^{0.05} \Phi_c(t) dt = \begin{bmatrix} t & t/6 + (1/36)e^{-6t} \\ 0 & -(1/6)e^{-6t} \end{bmatrix}_0^{0.05} = \begin{bmatrix} 0.05 & 0.001134 \\ 0 & 0.04320 \end{bmatrix}$$

$$B = \left[\int_0^{0.05} \Phi_c(t) dt \right] B_c = \begin{bmatrix} 0.001134 \\ 0.04320 \end{bmatrix}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 1 & 0.04320 \\ 0 & 0.7408 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.001134 \\ 0.04320 \end{bmatrix} e(k)$$

$$y(k) = [20 \quad 0] \underline{x}(k)$$



$$(f) \quad \frac{Y(z)}{E(z)} = \frac{0.02268z^{-1}(1 - 0.7408z^{-1}) + 0.03741z^{-2}}{1 - z^{-1} - 0.7408z^{-1} + 0.7408z^{-2}}$$

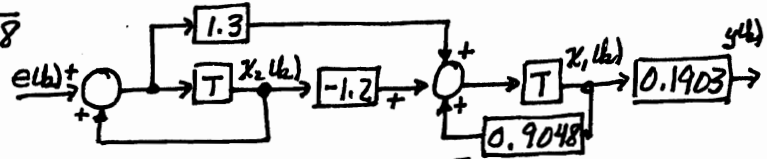
$$= \frac{0.02268z + 0.02061}{z^2 - 1.7408z + 0.7408}$$

(g)

```
num=[0 0 20];
den=[1 6 0];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,.05)
[n,d] = ss2tf(A,B,C,D)
pause
Ac = [0 1; 0 -6]; Bc = [0;1];
[A,B] = c2d(Ac,Bc,.05)
```

$$4-29. D(z) = 1.2 + \frac{0.1z}{z-1} = \frac{1.3z - 1.2}{z-1}$$

$$(a) G(z) = \frac{0.1903}{z - 0.9048}$$



$$\begin{aligned} x_1(k+1) &= 0.9408x_1(k) - 1.2x_2(k) + 1.3[e(k) + x_2(k)] \\ &= 0.9408x_1(k) + 0.1x_2(k) + 1.3e(k) \end{aligned}$$

$$x_2(k+1) = x_2(k) + e(k)$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0.9408 & 0.1 \\ 0 & 1 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1.3 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [0.1903 \quad 0] \underline{x}(k)$$

(b) Same filter simulation diagram as in (a).

From Problem 4-27(d):

$$\underline{x}(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} m(k)$$

$$m(k) = [-1.2 + 1.3]x_3(k) + 1.3e(k) = 0.1x_3(k) + 1.3e(k)$$

$$\therefore \begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0.05 \\ 0 & 1 & 0.1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0.65 \\ 1.3 \\ 1 \end{bmatrix} e(k)$$

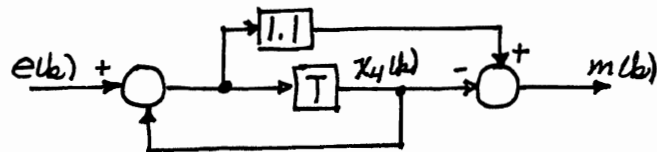
$$y[k] = [K/J \quad 0 \quad 0] \underline{x}(k)$$

(c)

```
nd = [1.3 -1.2]; dd = [1 -1];
ng = [0 0.1903]; dg = [1 -0.9048];
n = conv(nd,ng)
d = conv(dd,dg)
pause
A = [.9408 .1; 0 1]; B = [1.3; 1];
C = [0.1903 0]; D = 0;
[num,dem] = ss2tf(A,B,C,D)
```

```
nd = [1.3 -1.2]; dd = [1 -1];
ng = [0 0.5 0.5]; dg = [1 -2 1];
n = conv(nd,ng)
d = conv(dd,dg)
pause
A = [1 1 .05; 0 1 .1; 0 0 1]; B = [.65; 1.3; 1];
C = [1 0 0]; D = 0;
[num,dem] = ss2tf(A,B,C,D)
```

$$4-30. D(z) = 1 + \frac{0.1z}{z-1} = \frac{1.1z-1}{z-1}$$



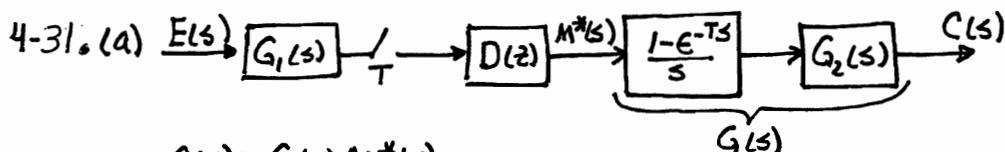
$$m(k) = -x_4(k) + 1.1[e(k) + x_4(k)] = 0.1x_4(k) + 1.1e(k)$$

$$\therefore \underline{x}(k+1) = A\underline{x}(k) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} [0.1x_4(k) + 1.1e(k)]$$

$$\text{and } x_4(k+1) = x_4(k) + e(k)$$

$$\therefore \begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \\ x_4(k+1) \end{bmatrix} = \begin{bmatrix} 0.9 & 1 & 2 & 0 \\ 0 & 0.9 & 0 & 0.1 \\ -1 & 0 & 0.5 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1.1 \\ 1.1 \\ 1 \end{bmatrix} e(k)$$

$$y(k) = [1 \quad 1.5 \quad 2.3 \quad 0] \underline{x}_e(k)$$



$$C(s) = G(s)M^*(s)$$

$$C(z) = G(z)M(z) = G(z)D(z)\overline{G_1E}(z)$$

(b) Cannot factor $E(z)$ from $\overline{G_1E}(z)$.

(c) Assume that $e(t)$ changes so slowly that the system can be accurately approximated by placing a sampler/data-hold in front of $G_1(s)$. Then

$$C(z) = G(z)D(z)z\left[\frac{1-e^{-Ts}}{s} \cdot G_1(s)\right]E(z).$$

$$4-32.(a) (1-0.5)\dot{x}(t) = -2x(t) + 3u(t)$$

$$\therefore \dot{x}(t) = -4x(t) + 6u(t)$$

$$y(t) = -4x(t) + 6u(t) + 4u(t) = -4x(t) + 10u(t)$$

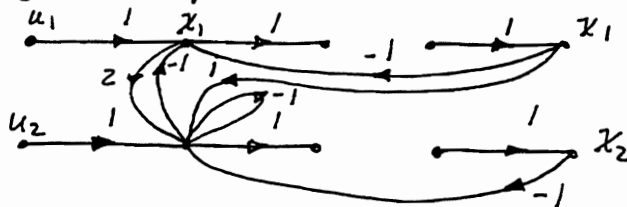
$$4-32.(b) \dot{\underline{x}} = \begin{bmatrix} 0 & 2 \\ -1 & -1 \end{bmatrix} \dot{\underline{x}} + \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \underline{u} = A_1 \dot{\underline{x}} + A_2 \underline{x} + B_1 \underline{u}$$

$$[I - A_1]^{-1} = \begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix}; [I - A_1]^{-1} A_2 = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

$$[I - A_1]^{-1} B_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix}; C_1 [I - A_1]^{-1} A_2 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \end{bmatrix}; C_1 [I - A_1]^{-1} B_1 = \begin{bmatrix} -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$\therefore \dot{\underline{x}}(t) = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \underline{x}(t) + \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix} \underline{u}(t)$$

$$y(t) = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \underline{x}(t) + \begin{bmatrix} -\frac{1}{4} & \frac{1}{4} \end{bmatrix} \underline{u}(t)$$



$$\Delta = 1 - (-2 - 1) = 4$$

$$\dot{x}_1 = \frac{-1(1+1)+2}{4} x_1 + \frac{-2}{4} x_2 + \frac{1(1+1)}{4} u_1 + \frac{2}{4} u_2$$

$$\dot{x}_2 = \frac{1+1}{4} x_1 + \frac{-1}{4} x_2 + \frac{-1}{4} u_1 + \frac{1}{4} u_2$$

$$y = \frac{1+1}{4} x_1 + \frac{-1}{4} x_2 + \frac{-1}{4} u_1 + \frac{1}{4} u_2$$